

Kiljae Lee

Statistics Ph.D. Candidate | Data-Centric AI | Explainable AI | LLM Evaluation | Uncertainty Quantification

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Education

Ph.D. in Statistics; Minor in Computer Science

The Ohio State University, Advisor: Dr. Yuan Zhang, GPA: 4.00/4.00

Aug 2022 – Expected Jun 2027

Columbus, OH

M.S. in Statistics

Korea University, Advisor: Dr. Taeryon Choi, GPA: 4.44/4.50

Mar 2020 – Aug 2021

Seoul, South Korea

B.S. in Statistics

Korea University, GPA: 4.14/4.50

Mar 2014 – Feb 2020

Seoul, South Korea

Research Signal

- **4 first-author papers at ICML, NeurIPS, and ICLR, including one co-first-author paper**, bringing statistical insight to data valuation, feature attribution, LLM evaluation, and uncertainty quantification in modern AI systems.
- Bridges **statistical theory and practical AI implementation**, translating Shapley-value and uncertainty quantification ideas into scalable methods for evaluating data quality, model behavior, and prediction uncertainty.
- Identifies **failure modes and adversarial vulnerabilities** in widely used data valuation methods, and develops robust frameworks for reliable data contribution measurement.

Selected Publications

Generalized Priority-Aware Shapley Value (NeurIPS 2026; first author)

Incorporates cyclic human preferences into LLM evaluation by modeling structured preferences as directed weighted priority graphs.

Priority-Aware Shapley Value (ICML 2026; first author)

Identifies data copying attacks in data valuation and introduces PASV, a robust data valuation framework that accounts for precedence constraints among data providers.

Faithful Group Shapley Value (NeurIPS 2025; ICML 2025 DataWorld Workshop; one of two co-first authors)

Identifies dataset splitting attacks in data valuation and proposes FGSV, a faithful data valuation method that defends against strategic manipulation.

Leave-One-Out Stable Conformal Prediction (ICLR 2025; first author)

Achieves $\geq 100\times$ speedups over standard conformal prediction methods by avoiding data splitting and iterative model fitting at the same time.

Research and Industry Experience

Data Scientist Intern

United Airlines

May 2026 – August 2026

Chicago, IL

- Built demand forecasting models that use customer search data as a proxy for latent total demand, improving willingness-to-pay estimation beyond observed booking behavior.

Graduate Research Assistant

The Ohio State University

Aug 2025 – Present

Columbus, OH

Statistical Consultant

The Ohio State University

May 2025 – Jul 2025

Columbus, OH

Additional Recognition and Service

- **ICML 2026 Gold Reviewer** (top 25% of reviewers); reviewer for ICLR (2025), ICML (2025, 2026), NeurIPS (2025, 2026), and AISTATS (2026).
- **Ransom & Marian Whitney Award for Research 2026**, The Ohio State University; one of two awardees, recognizing Ph.D. research independence, creativity, originality, and publication/application potential.
- **Best Paper Award Honorable Mention**, ICML 2025 DataWorld Workshop, for *Faithful Group Shapley Value*; ranked among the top 3 of 55 accepted submissions.

Technical Skills

Programming: Python, R, C++, SQL, Git

ML stack: PyTorch, Hugging Face Transformers, scikit-learn, NumPy/SciPy, pandas

LLM/Generative AI: Open-source {LLM/Diffusion} {fint-tuning/inference}, quantized models, PEFT, RLHF

Infrastructure: SLURM/HPC clusters, GPU computing, SSH/tmux